Supplementary Information

An engineered histidine's cost: substitution tolerance replicates across folding stability, variant effect prediction and human variation

Taner Karagöl[1,3,*], Alper Karagöl[2,3,*]

[1] McGovern Institute for Brain Research, Massachusetts Institute of Technology, Cambridge, MA, USA
[2] Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA, USA
[3] Istanbul University Istanbul Medical Faculty, Istanbul, Turkey
[*] To whom correspondence should be addressed.
Taner Karagöl, karagolt@mit.edu, taner.karagol@gmail.com, ORCID 0009-0005-1011-7661
Alper Karagöl, akaragol@mit.edu, alper.karagol@gmail.com, ORCID 0009-0001-7864-0732
Running Title: The cost of an engineered histidine

Table of Contents

1. Supplementary Table S1. Measured folding stability of each substitution, from the cDNA-proteolysis set with the curated and benchmark sets alongside Pg. 3
2. Supplementary Table S2. Rank agreement between readouts over the twelve substitutions Pg. 4
3. Supplementary Table S3. Cluster bootstrap over proteins, 1,000 resamples of each cohort Pg. 5
4. Supplementary Table S4. Functional stratification of the twelve substitutions in the transmembrane cohort, at an annotation window of zero residues Pg. 6
5. Supplementary Table S5. Matched Cliff delta inside deciles of leave-one-out positional constraint, transmembrane cohort Pg. 7
6. Supplementary Table S6. Designed against natural domains in the cDNA-proteolysis set Pg. 8
7. Supplementary Table S7. Variance in the per-protein effect of a substitution, separated into a between-substitution and a protein-by-substitution component Pg. 10
8. Supplementary Table S8. Predicting measured folding cost on held-out domains, from the substitution, from the position, and from both Pg. 11
9. Supplementary Table S9. Four checks on what the ordering is made of and how far it travels Pg. 12
10. Supplementary Table S10. The deposited structures the side-chain geometry is measured on Pg. 13
11. Supplementary Table S11. Each substitution against its baseline, with Benjamini-Hochberg correction across the twelve Pg. 14

12. Supplementary Table S12. Curated-variation pathogenic fractions with gene-clustered intervals Pg. 15
13. Supplementary Table S13. Shape overlap against measured folding cost, over every substitution class Pg. 16
14. Supplementary Figure S1. Functional stratification at a wider annotation window, and the FireProtDB flagged positions Pg. 17

Supplementary Table S1. Measured folding stability of each substitution, from the cDNA-proteolysis set with the curated and benchmark sets alongside.

Substitution	Dir.	Megascale n	Mean (kcal/mol)	Median	Cliff delta	FireProt n	FireProt Cliff	S669 n	S669 Cliff
E>Q	basic	1690	+0.193	+0.133	+0.318	30	+0.273	17	+0.168
K>Q	acidic	1551	+0.193	+0.146	+0.316	24	+0.260	-	-
Q>K	basic	580	+0.240	+0.118	+0.298	3	+0.437	-	-
Q>E	acidic	586	+0.266	+0.189	+0.274	5	+0.190	-	-
N>H	basic	683	+0.273	+0.149	+0.266	7	+0.379	-	-
D>N	basic	818	+0.272	+0.168	+0.261	31	+0.048	13	-0.224
N>D	acidic	673	+0.321	+0.210	+0.216	20	+0.315	-	-
E>K	basic	1665	+0.404	+0.308	+0.150	31	+0.217	5	-0.095
D>H	basic	815	+0.417	+0.295	+0.150	9	+0.101	3	-0.135
K>E	acidic	1535	+0.463	+0.373	+0.103	27	+0.026	-	-
H>N	acidic	262	+0.497	+0.379	+0.060	6	-0.284	-	-
H>D	acidic	253	+0.814	+0.730	-0.117	-	-	-	-
Baseline	-	112293	+0.694	+0.444	0	1839	0	402	0

Costs in kcal/mol, with a larger value meaning a less stable protein after the sign mapping described in the Methods. Cliff delta is computed against every single-nucleotide-reachable substitution in the same dataset, and a positive value means better tolerated than an average single-base substitution. Rows are ordered by the cDNA-proteolysis Cliff delta, where E>K at +0.15034 and D>H at +0.14958 round alike to three decimals and are separated at full precision. A dash marks a substitution with fewer than three measurements in that dataset. The baseline row gives each dataset's baseline count and, for the cDNA-proteolysis set, its mean and median cost.

Supplementary Table S2. Rank agreement between readouts over the twelve substitutions.

Comparison	Spearman rho	Edges compared	Permutation p	BH p across the eleven
AlphaMissense soluble against AlphaMissense transmembrane	+0.909	12	0.0001	0.0012
humsavar against AlphaMissense soluble Cliff delta	+0.846	12	0.0009	0.0038
humsavar against AlphaMissense transmembrane Cliff delta	+0.839	12	0.0010	0.0038
Megascale against humsavar tolerance	+0.783	12	0.0037	0.0102
Megascale against AlphaMissense soluble Cliff delta	+0.734	12	0.0084	0.0184
Megascale against FireProt Cliff delta	+0.682	11	0.0258	0.0472
Megascale against AlphaMissense transmembrane Cliff delta	+0.622	12	0.0340	0.0534
FireProt against AlphaMissense soluble Cliff delta	+0.482	11	0.138	0.1891
FireProt against humsavar tolerance	+0.455	11	0.163	0.1994
S669 against AlphaMissense soluble Cliff delta	+0.400	4	0.750	0.75
S669 against humsavar tolerance	+0.400	4	0.750	0.75

Spearman rho on the twelve substitutions, except where a dataset holds fewer than three measurements for some of them. Curated variation enters as tolerance, that is, as the negated pathogenic fraction, so that a positive rho means agreement in every row. Values are computed on the full-precision Cliff deltas; two pairs of substitutions round alike to three decimals and are separated at full precision. The permutation p is two sided over all orderings of one readout, exact where the number of substitutions allows it and over 200,000 random orderings otherwise. Benjamini-Hochberg is applied across all eleven rank agreements, which is every one computed and not only those that separate from chance.

Supplementary Table S3. Cluster bootstrap over proteins, 1,000 resamples of each cohort.

Substitution	Soluble Cliff	Soluble 95% interval	Soluble rank (95%)	TM Cliff	TM 95% interval	TM rank (95%)
Q>E	+0.413	+0.370 to +0.459	1 (1 to 1)	+0.494	+0.481 to +0.507	1 (1 to 1)
N>H	+0.233	+0.174 to +0.285	2 (2 to 4)	+0.198	+0.178 to +0.216	6 (5 to 7)
Q>K	+0.233	+0.183 to +0.287	3 (2 to 4)	+0.321	+0.304 to +0.337	2 (2 to 2)
K>Q	+0.191	+0.116 to +0.256	4 (2 to 6)	+0.226	+0.210 to +0.242	4 (4 to 5)
E>Q	+0.158	+0.122 to +0.190	5 (4 to 6)	+0.214	+0.200 to +0.226	5 (4 to 6)
H>N	+0.113	+0.024 to +0.207	6 (4 to 8)	+0.262	+0.244 to +0.281	3 (3 to 3)
D>N	+0.087	+0.048 to +0.130	7 (6 to 8)	+0.179	+0.165 to +0.192	7 (6 to 7)
N>D	+0.079	+0.029 to +0.125	8 (6 to 8)	+0.093	+0.072 to +0.113	8 (8 to 9)
E>K	-0.039	-0.080 to +0.001	9 (9 to 10)	+0.083	+0.065 to +0.102	9 (8 to 9)
K>E	-0.066	-0.139 to -0.003	10 (9 to 10)	+0.023	+0.002 to +0.043	10 (10 to 10)
H>D	-0.187	-0.262 to -0.106	11 (11 to 12)	-0.020	-0.040 to +0.001	11 (11 to 11)
D>H	-0.253	-0.292 to -0.217	12 (11 to 12)	-0.168	-0.181 to -0.156	12 (12 to 12)

Proteins are resampled with replacement and every substitution rescored, so the interval respects the fact that positions inside one protein are not independent. The rank column gives the pooled rank with the 2.5th and 97.5th percentiles of the bootstrap rank. Rows are ordered by the soluble Cliff delta.

Supplementary Table S4. Functional stratification of the twelve substitutions in the transmembrane cohort, at an annotation window of zero residues.

Substitution	n annotated	Matched Cliff, annotated (rank)	n unannotated	Matched Cliff, unannotated (rank)	% pathogenic, annotated	% pathogenic, unannotated
Q>E	70	+0.606 (1)	10745	+0.495 (1)	34.3	8.6
Q>K	68	+0.274 (2)	10747	+0.322 (2)	67.6	20.3
N>H	156	+0.204 (3)	11620	+0.203 (6)	71.8	26.2
E>Q	148	+0.157 (4)	15864	+0.216 (5)	72.3	20.6
N>D	157	+0.091 (5)	11612	+0.097 (8)	79.6	33.8
K>Q	128	+0.036 (6)	13635	+0.229 (4)	67.2	18.2
D>N	259	+0.030 (7)	12038	+0.190 (7)	76.4	27.1
E>K	148	-0.064 (8)	15890	+0.085 (9)	79.1	32.2
H>N	116	-0.065 (9)	6493	+0.273 (3)	77.6	22.0
K>E	129	-0.094 (10)	13640	+0.024 (10)	78.3	34.8
H>D	116	-0.302 (11)	6489	-0.012 (11)	85.3	37.9
D>H	258	-0.316 (12)	12031	-0.161 (12)	89.9	44.9
Baseline	13396	mean 0.7742	1769587	mean 0.4597	-	-

A position is annotated if UniProt marks it as an active site, a binding site or a site. Matched Cliff delta is computed against the baseline inside the same stratum, which removes the conservation difference between strata; the baseline row gives each stratum's size and mean AlphaMissense score. Ranks run from 1 to 12 within each stratum. Pathogenic percentages are the share of substitutions AlphaMissense calls likely pathogenic. Rows are ordered by matched Cliff delta at annotated positions.

Supplementary Table S5. Matched Cliff delta inside deciles of leave-one-out positional constraint, transmembrane cohort.

Substitution	D1 least	D2	D3	D4	D5	D6	D7	D8	D9	D10 most
Q>E	+0.45	+0.63	+0.65	+0.63	+0.55	+0.55	+0.57	+0.62	+0.66	+0.78
Q>K	+0.20	+0.44	+0.46	+0.37	+0.20	+0.05	-0.05	-0.07	-0.17	+0.01
H>N	+0.22	+0.16	+0.12	+0.12	+0.15	+0.14	+0.24	+0.30	+0.33	+0.19
K>Q	+0.02	+0.34	+0.49	+0.52	+0.47	+0.49	+0.52	+0.50	+0.33	-0.02
E>Q	+0.13	+0.30	+0.40	+0.40	+0.32	+0.35	+0.42	+0.50	+0.56	+0.46
N>H	+0.38	+0.43	+0.41	+0.36	+0.27	+0.25	+0.28	+0.26	+0.25	+0.15
D>N	+0.22	+0.46	+0.54	+0.53	+0.48	+0.50	+0.56	+0.58	+0.63	+0.35
N>D	+0.19	+0.18	+0.12	+0.09	+0.02	-0.05	-0.05	-0.05	-0.08	-0.14
E>K	+0.16	+0.36	+0.28	+0.15	-0.04	-0.10	-0.09	-0.06	-0.01	-0.09
K>E	+0.29	+0.43	+0.28	+0.10	-0.04	-0.11	-0.15	-0.28	-0.44	-0.49
H>D	-0.35	-0.58	-0.62	-0.63	-0.58	-0.57	-0.51	-0.45	-0.40	-0.43
D>H	-0.49	-0.45	-0.43	-0.39	-0.36	-0.34	-0.29	-0.24	-0.15	-0.30

A position's constraint is the mean AlphaMissense score of every substitution at it except the one being scored, so the substitution under test does not contribute to the score that strata it. D1 holds the least constrained tenth of positions and D10 the most. Each value is a Cliff delta against the baseline inside the same decile. Rows are ordered by the pooled Cliff delta.

Supplementary Table S6. Designed against natural domains in the cDNA-proteolysis set.

Substitution	Natural n	Natural mean	Natural Cliff (95%)	Designed n	Designed mean	Designed Cliff (95%)	Difference
E>Q	921	+0.153	+0.353 (+0.320 to +0.387)	769	+0.240	+0.279 (+0.202 to +0.346)	-0.074
Q>K	440	+0.225	+0.316 (+0.268 to +0.362)	140	+0.288	+0.233 (+0.129 to +0.325)	-0.084
K>Q	936	+0.196	+0.311 (+0.278 to +0.344)	615	+0.188	+0.331 (+0.268 to +0.386)	+0.021
N>H	473	+0.233	+0.310 (+0.269 to +0.350)	210	+0.362	+0.152 (+0.069 to +0.234)	-0.158
Q>E	445	+0.255	+0.292 (+0.243 to +0.334)	141	+0.303	+0.203 (+0.109 to +0.297)	-0.089
N>D	463	+0.288	+0.263 (+0.210 to +0.316)	210	+0.394	+0.104 (+0.036 to +0.172)	-0.159
D>N	612	+0.278	+0.258 (+0.214 to +0.301)	206	+0.252	+0.270 (+0.186 to +0.346)	+0.012
E>K	904	+0.410	+0.164 (+0.116 to +0.215)	761	+0.396	+0.133 (+0.053 to +0.208)	-0.031
D>H	609	+0.437	+0.148 (+0.098 to +0.193)	206	+0.356	+0.164 (+0.083 to +0.240)	+0.016
H>N	170	+0.459	+0.120 (+0.044 to +0.192)	92	+0.566	-0.066 (-0.157 to +0.025)	-0.186
K>E	924	+0.518	+0.082 (+0.034 to +0.130)	611	+0.380	+0.142 (+0.071 to +0.205)	+0.060
H>D	161	+0.851	-0.086 (-0.183 to	92	+0.750	-0.191 (-0.288 to	-0.106

			+0.015)			-0.098)	
Y>H	383	+1.161	-0.298 (-0.360 to -0.235)	113	+1.020	-0.393 (-0.473 to -0.314)	-0.095

[185],267 measurements over 189 natural domains and 85,964 over 109 designed ones, each class scored against its own single-nucleotide baseline, which is 77,164 substitutions at a mean cost of +0.731 kcal/mol for the natural class and 35,129 at +0.612 for the designed one. Intervals are from resampling domains within the class. Y>H is listed because the de novo exemption proposed for it is what this comparison tests. Rows are ordered by the natural Cliff delta.

Supplementary Table S7. Variance in the per-protein effect of a substitution, separated into a between-substitution and a protein-by-substitution component.

Set	Readout	Between-substitution	Protein-by-substitution	Share from the substitution	Transfer R squared	Cells
All 361 substitution classes	measured folding stability	0.2949	0.0955	76%	0.48	44,557
The twelve	transmembrane cohort	0.0077	0.0061	56%	0.36	5,790
The twelve	measured folding stability	0.0256	0.0631	29%	0.08	1,811

Components are fitted by restricted maximum likelihood with the sampling variance of each cell estimated from the spread inside it and removed. The share is the between-substitution component over the sum of the two. Transferability is leave-one-protein-out: predicting a protein's value for a substitution from that same substitution measured in every other protein. Variances are in the units of each readout, kcal/mol squared for measured folding stability and squared pathogenicity score for the cohort, so they are comparable within a row, not between rows.

Supplementary Table S8. Predicting measured folding cost on held-out domains, from the substitution, from the position, and from both.

Predictor	All classes, Spearman	All classes, R squared	The twelve, Spearman	The twelve, R squared
Substitution identity alone	+0.479	0.264	+0.168	0.042
Position signal alone	+0.742	0.570	+0.705	0.512
Both together	+0.779	0.653	+0.722	0.555
What the substitution adds on top of the position		+0.084		+0.043
What the position adds on top of the substitution		+0.390		+0.513

Five folds split by domain. The position signal is the mean measured cost of every other substitution at the same position, which bounds what any position-specific predictor could extract from that position. Values are averaged across folds, over 271,181 held-out measurements for all classes and 11,111 for the twelve.

Supplementary Table S9. Four checks on what the ordering is made of and how far it travels.

Check	Quantity	Result
Side-chain chemistry, the twelve	R squared, charge magnitude alone	0.019
	R squared, volume and hydropathy	0.458
	R squared, all three descriptors	0.635
Side-chain chemistry, all 361 classes	R squared, all three descriptors	0.128
Baseline	rank agreement, every substitution	+1.000
	rank agreement, charge-matched pool	+0.902
Fold, designed constructs	rank agreement, all-alpha against mixed	+0.188
Mutational access	median transition to transversion ratio	3.04

Regressions are on the per-substitution Cliff delta in the cDNA-proteolysis set, using the magnitudes of the charge, volume and hydropathy changes. Rank agreement is Spearman's rho against the ordering computed on the single-nucleotide-reachable baseline. The topology comparison uses the designed constructs only, which are named by fold. The transition ratio is the count of curated human variants on the transition route over the count on the transversion route leaving the same source residue, which holds the number of available positions fixed.

Supplementary Table S10. The deposited structures the side-chain geometry is measured on.

Deposited structures used for the side-chain geometry
1A32, 1AOY, 1E0L, 1E6H, 1ENH, 1F0M
1GJS, 1GL5, 1GYZ, 1H8K, 1H92, 1I2T
1I6C, 1IFY, 1JIC, 1K1V, 1LP1, 1O6X
1OPS, 1ORC, 1PSE, 1PV0, 1PWT, 1QKH
1QKX, 1QLY, 1QP2, 1R69, 1SRM, 1TG0
1TUC, 1TUD, 1UFM, 1URF, 1UZC, 1V1C
1VII, 1W4F, 1W4G, 1W4H, 1WCL, 1WR4
1Y0M, 1YP5, 1YRF, 1YU5, 1ZHC, 1ZLM
2AMI, 2B88, 2B89, 2BTH, 2BTT, 2CDT
2CJJ, 2D1U, 2EXD, 2GP8, 2HBB, 2HDZ
2JN4, 2JT1, 2JTV, 2JVD, 2JVG, 2JWS
2JWT, 2JZ2, 2K1B, 2K28, 2K2A, 2K5H
2K5N, 2K5P, 2K9D, 2KCF, 2KCM, 2KFV
2KGT, 2KR3, 2KRU, 2KVS, 2KVT, 2KVV
2KWH, 2KXD, 2KZJ, 2L09, 2L2D, 2L2P
2L33, 2L6Q, 2L7F, 2L7M, 2L9R, 2LC2
2LCL, 2LDM, 2LGW, 2LHC, 2LHR, 2LJ3
2LO1, 2LP5, 2LQK, 2LUM, 2LVN, 2LXE
2LYQ, 2LYR, 2M0C, 2M0Y, 2M2J, 2M2L
2M5A, 2M6Y, 2M8E, 2M8I, 2M8J, 2M8U
2M9E, 2M9F, 2M9I, 2MA4, 2MC5, 2MCH
2MCK, 2MH8, 2MI6, 2MKY, 2MW9, 2MWA
2MWB, 2MXD, 2MYX, 2N4R, 2N88, 2O2W
2OCH, 2OP7, 2QFF, 2RJV, 2RRT, 2RRU
2RU9, 2WNM, 2WQG, 2WXC, 2YSB, 2YSF
3CQT, 3DKM, 3I35, 3L1X, 3MYC, 3V1A
3ZGK, 4C26, 4G3O, 4HCK, 4UZW, 4UZX
5AHT, 5JRT, 5KPH, 5LXJ, 5OAO, 5UBS
5UCE, 5UP5, 5UYO, 5VNT, 5Z2S, 6ACV
6EWS, 6EWT, 6EWU, 6FVC, 6IVS, 6IWS
6M3N, 6NMW, 6OBK, 6SCW, 6SOW, 6YSE
7BPM, 7JJK

The 188 natural domains of the cDNA-proteolysis set, named in that set by their four-character Protein Data Bank identifier. The largest chain of each entry was taken, which keeps the deposited domain where the entry is a complex and takes one copy where every chain is a copy. All 188 were parsed; none was skipped.

Supplementary Table S11. Each substitution against its baseline, with Benjamini-Hochberg correction across the twelve.

Substitution	Soluble Cliff	Soluble 95%	Soluble BH p	TM Cliff	TM 95%	TM BH p
Q>E	+0.449	+0.395 to +0.510	0.0002	+0.521	+0.509 to +0.533	0.0001
K>Q	+0.229	+0.145 to +0.305	0.0002	+0.257	+0.242 to +0.272	0.0001
E>Q	+0.187	+0.132 to +0.239	0.0002	+0.245	+0.235 to +0.256	0.0001
Q>K	+0.241	+0.172 to +0.322	0.0002	+0.344	+0.328 to +0.359	0.0001
N>D	+0.112	+0.054 to +0.164	0.0003	+0.121	+0.103 to +0.140	0.0001
N>H	+0.286	+0.208 to +0.359	0.0002	+0.230	+0.213 to +0.247	0.0001
H>N	+0.141	+0.040 to +0.248	0.0067	+0.274	+0.255 to +0.293	0.0001
K>E	-0.050	-0.124 to +0.030	0.2096	+0.054	+0.034 to +0.074	0.0001
D>N	+0.124	+0.080 to +0.172	0.0002	+0.214	+0.202 to +0.227	0.0001
E>K	-0.041	-0.093 to +0.017	0.1800	+0.117	+0.099 to +0.135	0.0001
H>D	-0.216	-0.302 to -0.121	0.0002	-0.023	-0.044 to -0.003	0.0266
D>H	-0.254	-0.305 to -0.198	0.0002	-0.158	-0.169 to -0.147	0.0001
N>H minus D>H	+0.540	+0.449 to +0.638	0.0001	+0.388	+0.367 to +0.410	0.0001

Cliff delta and its interval are from 10,000 cluster bootstrap resamples over proteins, so the value is the protein-averaged effect and differs slightly from the pooled value of Table 2. The p is two sided and corrected across the twelve inside each cohort. E>K and K>E are the two substitutions that do not separate from the baseline in the soluble cohort.

Supplementary Table S12. Curated-variation pathogenic fractions with gene-clustered intervals.

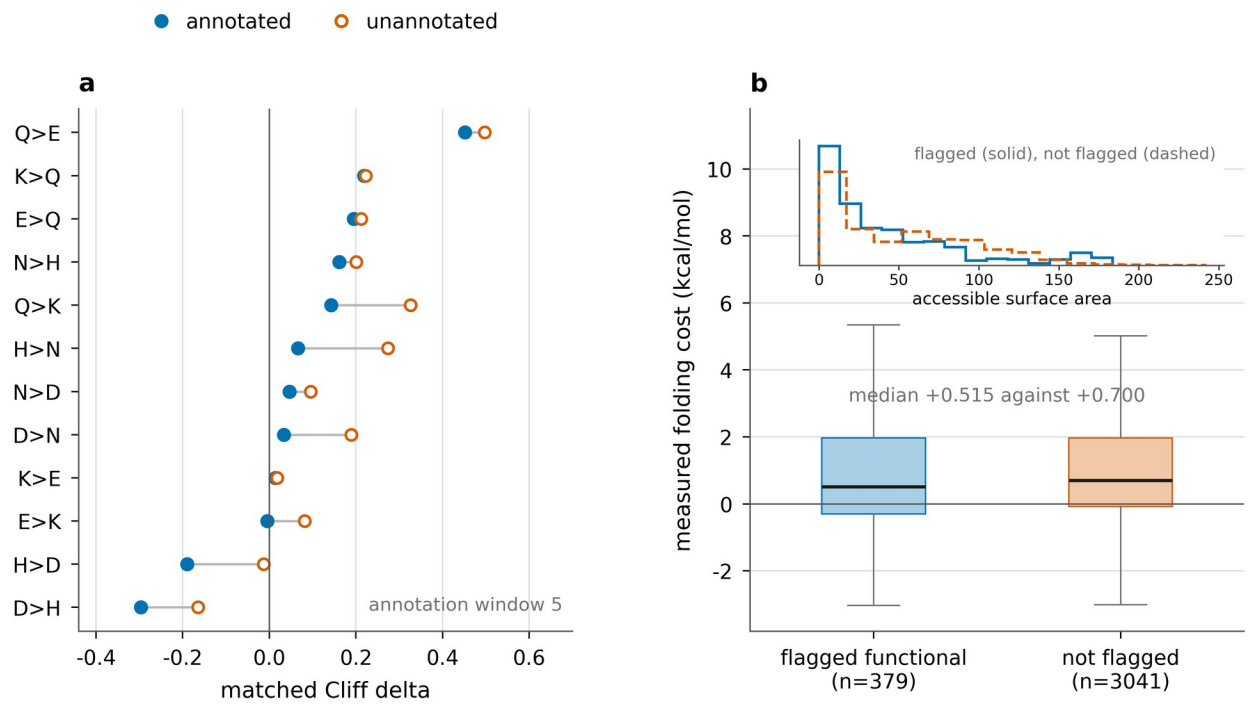
Substitution	Variants	Genes	Pathogenic %	Binomial 95%	Gene-clustered 95%
Q>E	260	244	24.6	19.8 to 30.2	19.3 to 30.7
K>Q	198	181	25.3	19.7 to 31.7	18.6 to 31.7
E>Q	398	358	29.1	24.9 to 33.8	24.3 to 34.3
Q>K	213	200	34.3	28.2 to 40.9	27.8 to 41.2
N>D	383	352	34.5	29.9 to 39.4	29.3 to 39.5
N>H	159	151	35.8	28.8 to 43.6	28.4 to 43.6
H>N	111	106	36.0	27.7 to 45.3	26.7 to 45.9
K>E	636	551	39.3	35.6 to 43.2	34.9 to 43.8
D>N	1109	851	45.5	42.6 to 48.5	41.9 to 49.1
E>K	1698	1195	52.0	49.6 to 54.4	48.4 to 55.5
H>D	121	114	52.9	44.0 to 61.6	43.9 to 62.1
D>H	350	300	55.4	50.2 to 60.5	49.3 to 61.3
D>H minus N>H			+19.6		+10.7 to +28.7

The binomial interval is the Wilson interval on the variant count. The gene-clustered interval resamples the 12,692 genes with replacement 2,000 times. The two differ by between 3 and 50 per cent in width, because the variants are spread at close to one per gene.

Supplementary Table S13. Shape overlap against measured folding cost, over every substitution class.

Quantity	Value	n	p
overlap vs Cliff delta, ordered classes	+0.313	306	0.00005
overlap vs mean Cliff delta, residue pairs	+0.528	153	0.00005
the same, change in heavy atoms held constant	+0.455	153	-
the same, change in reach held constant	+0.509	153	-
|change in heavy atoms| vs mean Cliff delta	-0.313	153	-
|change in reach| vs mean Cliff delta	-0.232	153	-
overlap range, all classes	0.100 to 0.857	306	-
overlap range, the twelve	0.501 to 0.846	12	-
residue types measured	19	10967	-
residue types used in the test	18	10964	-

Overlap is symmetric, so a residue pair carries one overlap and the mean of the two directional costs, and the 153 pairs are the independent rows. The partial correlations hold the change in heavy atoms and the change in reach constant in turn. The twelve occupy a narrow band of the full overlap range, which is why the measure has little to resolve inside them.



Supplementary Figure S1. Functional stratification at a wider annotation window, and the FireProtDB flagged positions.

(a) Matched Cliff delta per substitution at annotated and at unannotated positions in the transmembrane cohort at an annotation window of five residues, ordered by the annotated value. Figure 3d gives the same comparison at a window of zero. (b) Measured folding cost at FireProtDB's flagged and unflagged positions, with the accessible surface area distribution of each group inset.